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A Multi-Modal Modified-Feedback Self-Paced BCI To Control the Gait of an Avatar

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Abstract

Brain-computer interfaces (BCI) have been used to control the gait of a virtual self-avatar with a proposed application in the field of gait rehabilitation. Some limitations of existing systems are: 1- some systems use mental imagery (MI) of movements other than gait; 2- most systems allow the user to take single steps or to walk but do not allow both; 3- most function in a single BCI mode (cue-paced or self-paced). **Objective**: The objective of this study was to develop a high performance multi-modal BCI to control single steps and forward walking of an immersive virtual reality avatar. **Approach**:This system used MI of these actions, in cue-paced and self-paced modes. Twenty healthy participants participated in this study, which was comprised of 4 sessions across 4 different days. They were cued to imagine a single step forward with their right or left foot, or to imagine walking forward. They were instructed to reach a target by using the MI of multiple steps (self-paced switch-control mode) or by maintaining MI of forward walking (continuous-control mode). The movement of the avatar was controlled by two calibrated RLDA classifiers that used the μ power spectral density (PSD) over the foot area of the motor cortex as a feature. The classifiers were retrained after every session. For a subset of the trials, positive modified feedback was presented to half of the participants, where the avatar moved correctly regardless of the classification of the participants' MI. The performance of the BCI was computed on each day, using different control modes**. Main results:** All participants were able to operate the BCI. Their average offline performance, after retraining the classifiers was 86.0 ± 6.1%, showing that the recalibration of the classifiers enhanced the offline performance of the BCI ($p < 0.01$). The average online performance was 85.9 ± 8.4% showing that modified feedback enhanced BCI performance ($p = 0.001$). The average performance was 83% at self-paced switch control and 92% at continuous control mode. **Significance**: This study reports on a first BCI to use motor imagery of the lower limbs in order to control the gait of an avatar with different control modes and different control commands (single steps or forward walking). BCI performance is increased in a novel way by cmobining three different performance enhancement techniques, resulting in a single high performance and multi-modal BCI system. This study also showed that the improvements due to the effects of mofified feedback lasted for more than one sesson.

Keywords: Brain-computer interface, Virtual reality, EEG, Classification, Avatar, Gait Rehabilitation

1. Introduction

A brain-computer interface (BCI) is a system that measures brain activity of an intention to do something and converts it into a control command that replaces, restores, enhances, supplements or improves natural brain activity output [1-3]. This control command has been used to control: assistive exoskeletons [4], self-navigation in virtual reality (VR) [5, 6] or the movements of a virtual self-avatar [7-9]. Such uses of BCI technology have found increasingly widespread applications in the field of neurorehabilitation [10-12] where it has been used to control the ambulation of a virtual self-avatar in VR in a Spinal Cord Injury (SCI) patient [7] and in post-stroke individuals [13-17]. In this type of BCI, users imagine the movement of a specific limb of their body and this motor imagery (MI) is detected by the BCI and translated into control commands that result in an action in VR or in movements of their avatar [18].

When this occurs, an illusion of embodiment of the virtual body can be induced [19-21]. Embodiment is the perceptual illusion whereby one perceives a virtual body, in part or in whole, as being their own [22]. The induction of such an illusion is important in MI-BCIs, where reaching a high level of both BCI performance and embodiment are inter-connected. To reach a high level in one of them, the second must also be reached to a high level [23, 24].

In addition to its role in inducing embodiment, MI of an intended limb movement also induces changes in μ (8–12 Hz) and β (16-30 Hz) rhythms over the corresponding sub-region of the sensorimotor cortex [25, 26]. Previous studies have shown that providing virtual visual feedback corresponding to the MI of intended movements can help patients can gradually recover from impairment through neuroplasticity [10, 26-31].

However, the benefits of using MI of the lower limbs during gait have not been as widely shown as they have been for upper limb movements, partly because of the complexity of the neural and biomechanical control of gait [32]. Hence, several studies have used MI of upper limb movements in a BCI to control the feedback of the navigation or lower limbs of an avatar [33-38] or of an exoskeleton [37]. For example, Hazrati and Hofmann [38] used the signals of MI of right/left hand movements to control the left/right navigation of an avatar. The same BCI with the same classification paradigm and mapping was used in [39], but this time with the additional possibility of self-paced navigation of the avatar. In similar work, the Linear Discriminant analysis (LDA) classifier output of IM of the manipulation of a cube was used in a BCI to control the navigation of an avatar in a rehabilitation room [36]. Such methods have been shown to allow a user to control the gait of an avatar but the fact that the imagined movement is different than the produced movement of the avatar prevents it's use in gait rehabilitation. Indeed, such a BCI would not allow the user to benefit from the neural plasticity properties of MI training, which is a crucial part in rehabilitation and restoring or enhancing motor functions [16, 40]. Moreover, performing MI of one movement and receiving visual feedback of another movement, sometimes even from a different limb, would not be conducive to the feeling of embodiment over the virtual avatar. This would therefore have a detrimental effect on BCI performance [41, 42].

To overcome this limitation, many studies have focused on the EEG signatures of gait, such as left and right foot discrimination [43, 44], gait initiation and termination in order to move forward and stop [45] and normal gait cycles [46]. They found that brain areas employed by these controls are lateralized (for steps) [43, 47, 48] and centralized (for walking) [45]. However, few studies have used MI of the lower-limbs in a BCI system that controls walking feedback [7, 49, 50]. To our knowledge, only two studies have used MI of the lower-limbs in a BCI system to control the feedback of taking steps [51]. For example, Donati et al. [51] found that using long-term training of paralyzed patients with a lower-limb MI-BCI to control the left and right steps of a virtual self-avatar, and later of a lower-limb exoskeleton, lead to partial neurological recovery. Such studies show promising results but they limit patients to only one type of command for walking (individual left and right steps) without allowing patients to progress to using the imagination of walking in normal gait cycles [7, 52].

On the other hand, a user can control BCIs in different modes. In cue-paced BCIs, the user is cued as to when to start producing the MI and the EEG signal has to be analyzed in predefined time windows. In contrast, self-paced BCIs continuously analyze EEG data, allowing the user to produce the specific mental pattern whenever he/she wishes. These self-paced BCIs can be further categorized in two modes. The first uses a brain switch control where the feedback is provided only once after the classification (switch-control mode) [53]. The second uses a continuous control , where the feedback is provided continuously for as long as the MI is maintained by the user (continuous-control mode) [54]. Each BCI control mode has its specific benefits to the rehabilitation process, depending on the training program of the rehabilitation process, the current phase within the training program of the rehabilitation process, the level of pathology and the progress of the user within his training program of the rehabilitation process. For example, the switch control mode is most suitable to control initiation of individual steps, which requires an on/off control strategy. But with the progression of the neurorehabilitation process, the continuous control mode becomes essential to be able to control the gait as a succession of left and right step motor commands at same time. Furthermore, the current gait MI-BCIs lack the possibility to enable the user to control BCIs in different modes at the same time [55].

Therefore, when designing a BCI for gait rehabilitation, a proposed way to overcome the aforementioned limitations is to allow the use of MI of left/right steps and of forward walking at the same time, and to map these signals to control the gait of an avatar in different modes (cue-pace, self-paced switch, self-paced continuous). Given the advantages of the different BCI-control modes for neurorehabilitation [56], the concurrent implementation of all of them would allow to accommodate different rehabilitation programs and patient progression within them. However, the combination of more control options and several modes in a single system would result in diminished performance, which is already low in lower-limb MI-BCIs, in comparison to upper-limb MI-BCIs [3, 55]. This lower performance is particularly problematic for rehabilitation applications because receiving feedback that is incongruent with the imagined movement would diminish embodiment and be detrimental to achieving neural plasticity benefits [16].

To overcome performance limitations, three enhancement techniques have been used in previous studies. The first consists of using a co-adaptive sequential experimental protocol to train users over different control modes [57]. Experimenters gradually increase the task difficulty in order to help the user stay engaged in the learning process, adapt to this process and tune his/her performance [58, 59]. They take into account the performance of the user at every stage of the training, using the method described by Tariq et al. [55]. Using this approach, the brain also adapts to the classifier periodically, thus getting the maximum benefits of the neural changes induced throughout the BCI training phases. For example, Allison et al. [60] shaped the cursor-control BCI training gradually from switch-control mode to continuous-control mode. In two different studies, it took 6 – 10 days [61] to master the control of the BCI [62]. Scherer et al. [8] trained users over a period of 3 days to control navigation in VR through a MI-BCI.

The second enhancement technique is the use of modified feedback. Feedback that is either positively or negatively modified has been shown to result in an adaptation on the part of participants, with performance enhanced up to 10% [63]. The BCI of Luu et al. [64] was used to control the gait of the left foot of a virtual avatar. After and initial training period, asymmetric gait patterns of the avatar were introduced over 8 days, resulting in neural and gait adaptation in participants. Similarly, Gonzalez-Franco et al. [65] and Alimardani et al. [66] reported that positive feedback had a greater learning effect on MI-BCI performance and the ownership illusion. This is also consistent with the results of Lotte et al. [67].

The third enhancement technique is used when we have a BCI training of multiple sessions, and it is to retrain the classifier after every session [68, 69]. This technique reduces the problem of the long calibration time that is required for BCIs, due to the large amount of calibration data that is required. For example, Sun and Zhang [70] trained users to control a cue-based BCI over 3 sessions, and adaptively updated the classifier based on new data from each session. The classification results were improved by an average of 10% between session 1 and session 3, versus a 2% improvement when using non-retrained classifiers. The same improvements were reported by Shenoy et al. [71] , Acqualagna et al. [72] and Llera et al. [73].

The main objective of this study was to design and evaluate a BCI for gait rehabilitation that integrates MI of left/right steps and forward walking at the same time, mapping these MI signals to control the gait of an avatar in immersive VR. This BCI had to allow control in cue-based mode and in the different self-paced modes (switch and continuous). We hypothesized that participants would be able to operate this BCI in all modes, with a better performance in cue-paced mode than in self-paced

and for self-paced more. We further hypothesized that performance in switch-control mode would be higher than in continuous-control mode. The secondary objective of this study was to investigate the increased performance of the BCI by integrating the three aforementioned performance enhancement techniques. We hypothesized that each of these techniques would contribute to increasing the overall performance of the BCI.

2. Materials and Methods

2.1 Data acquisition and recording

EEG signals were recorded with a Smart BCI system (Mitsar, Russia). This system consists of an EEG cap with 19 Ag/Cl cup

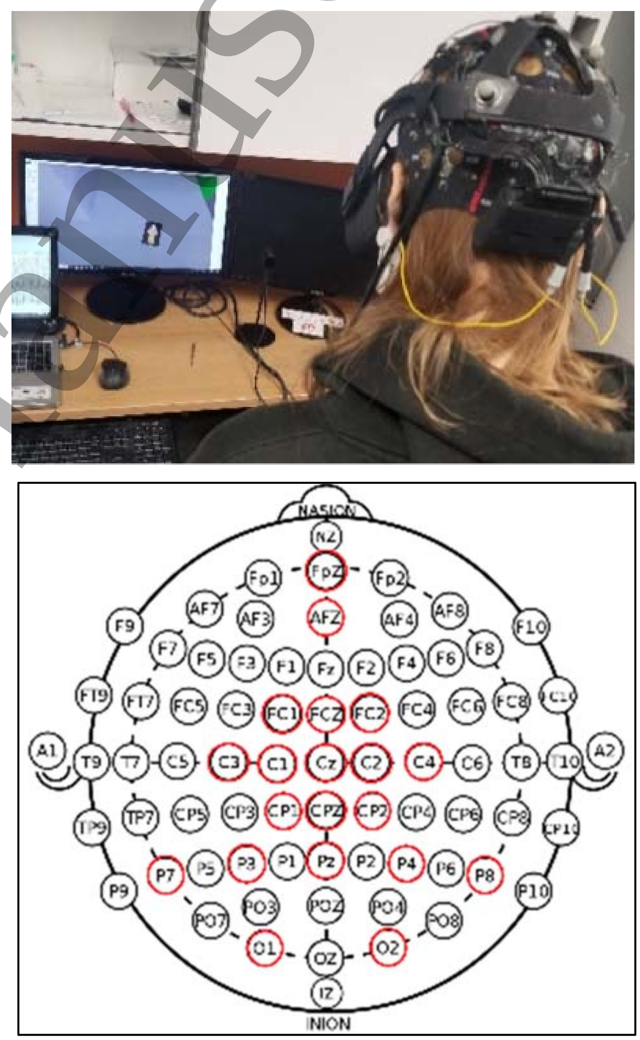

Figure 1: Top: a participant controls the BCI by imagining a right step while wearing the HMD and EEG cap. Bottom: EEG-electrode placement design for this study. The positions of the electrodes are identified by red circles.

electrodes. The electrode positions were set to cover the main regions of interest (Figure 1) in this study. Thus, most of the electrodes were sitting over the pre-motor, motor and parietal areas. The EEG electrodes were placed as stated by the 10-20 system and applied to the participant’s scalp with conductive gel. The electrodes cover the central area under electrodes C1, C2, C3, C4 (lateralised feet control) [43, 47, 48] and Cz (forward walk control) [45], the central frontal area under electrodes FCz, FC1 and FC2 (movement planning) [45], the frontal area (FPz) [74] the central-parietal area under electrodes CPz, CP1, and CP2 (spatial navigation and feet control) [45], the parietal area under electrodes Pz, P3, P4, P7 and P8 (spatial navigation and sense of presence) [43, 44] and the occipital area (O1 and O2). The AFz electrode was used as ground, and an ear clips on each ear were used for reference. Since the HMD was positioned right over the EEG cap, special care was taken when installing it, in order to not displace the electrodes.

When controlling the avatar via the EEG-based BCI, the EEG signals where streamed to EEGStudio software (Mitsar, Russia) via Bluetooth, which in turn streamed them in real-time via an API to MATLAB. These signals were analyzed online, then control signals were formed. The TCI-IP protocol was used to send these control commands to the virtual avatar, in real-time.

An Oculus Rift HMD was used in this experiment in order to immerse the participant in a VE. This VE, developed in Unity 3D (Unity Technologies, USA), was an infinite virtual corridor that was a replica of the hallway in our research center (Figure 2). To increase the perception of forward progress within the corridor, horizontal lines were added to the floor texture, increasing the optical flow. A generic virtual self-avatar was created using MakeHuman. The size of the avatar was set to be proportional to the size of the VE. The height of the camera was calibrated according to the height of the participant in order to align the visual perspective with the positions of the avatar's eyes. The avatar was animated using a generic human walking animation obtained from Mixamo (Adobe, USA).
When the avatar initiated a forward step or forward walking within the virtual corridor, a realistic optical flow of the VE was produced.

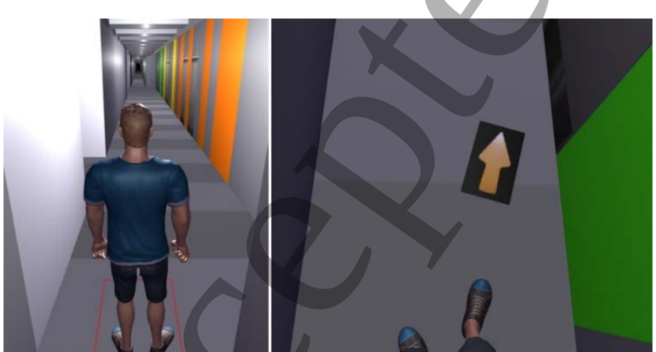

Figure 2: The VE that was used in this study. The left image shows the virtual avatar displayed from a 3PP; the right image shows the avatar displayed to the participant from 1PP in their HMD. It also shows the arrow cue for a right step.

2.2 Experimental Protocol

Twenty participants that fit the following inclusion criteria were recruited for this study: aged between 18 and 45 years old, in good general health, not taking any medication that acts on the central nervous system, with good vision (with or without glasses), and not suffering from motion sickness. These participants (12 women, 8 men; aged 26.7 ± 6.1 years old) were recruited through our research laboratory's newsletter emails. The experiment was approved by the Research Ethics Committees of the University of Montreal Hospital Center (CHUM) and of Ecole de technologie superieure (ETS), project ID number 16.170. All recruited participants signed an informed consent form prior to their participation in the study.

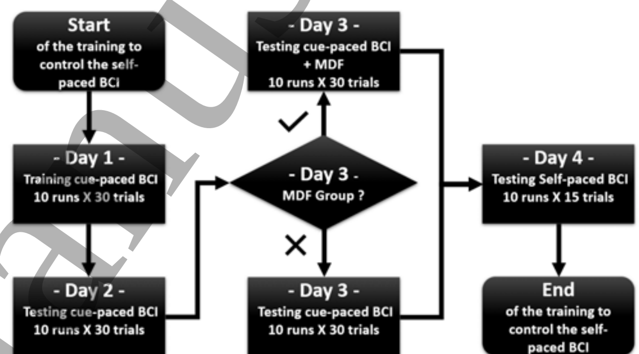

Figure 3: General experimental design of the study

The study was conducted over 4 consecutive days (1 session per day), in a co-adaptive sequential training, in order to control a BCI in a self-paced paradigm (Figure 3). During the experiment, participants were instructed to wear a head-mounted display (HMD) and to always focus on the lower limbs of their self-avatar, displayed in this device from a first-person perspective (1PP). They were instructed to "imagine" three different types of movements. The experiment lasted approximately 1.5 hours per session, including setup time and regular breaks.

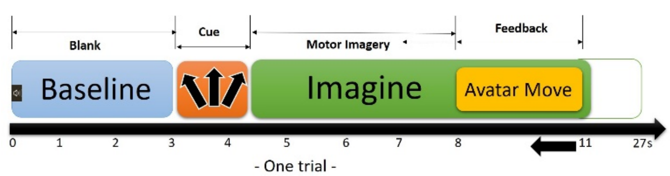

Figure 4: Experimental design of one trial, showing the total duration of 11 s for each trial of days 1 to 3, and the total duration of 27 s for each trial of day 4

Training in days 1 through 3 included 10 runs, with 30 trials each, for a total of 300 trials. The training in day 4 included 10 runs of 15 trials, for a total of 150 trials (Figure 2). Each trial lasted approximatively 11 seconds and started with an auditory cue (a beep) to indicate the beginning of a trial. It was followed by a baseline EEG acquisition that lasted three-seconds. Then, for a period of 1.25 s, the participant was shown a yellow arrow on the virtual floor, 152 cm in front of his feet. This arrow was used to cue the three different types of movements the participant had to imagine during the experiment. When the arrow was pointing to the left, the participant had to imagine a step with the left foot, when the arrow was pointing to the right, the participant had to imagine a step with the right foot and when the arrow was pointing straight forward, the participant had to imagine walking forward. Following MI of the action, the participant received feedback in the form of an action from his/her self-avatar (Figure 4).

2.2.1 Cue-Paced BCI design

On day 1, EEG was acquired during the MI the participants performed of the cued actions, without them physically moving. The avatar performed the cued movements after the cue disappeared, but the actions of the avatar were independent of the recorded brain activity. The main goal of this session was to acquire the signals related to the MI of the participants and train two classifiers. At the end of this session, signals were analyzed, and the two classifiers were trained: a classifier with two possible classes: walking forward or no movement (Cl1), and a classifier with three possible classes: right step, left step or no movement (Cl2).

On day 2, the previously trained classifiers were used in the same paradigm as on day 1, but this time the avatar performed the movements according to the classification of the brain activity signal. The classifier output was updated every 200ms. In order to change from the no-movement to the movement state, and to prevent rapid changes, the subject had to accumulate more than a "selection time" with the correct movement selected [54, 75]. The selection time is the number of successive correct classifier outputs resulting from MI of a movement. The selection time was set to three consecutive outputs, i.e. 600 ms.

If the MI was held for less than this selection time, the state remained in no-movement. Similarly, if the MI output classification was for the incorrect class, the state remained in no-movement.

2.2.2 Modified Feedback Cue-Paced BCI design

On day 3, in order to enhance the BCI performance, two techniques were used: 1- the two classifiers were updated and retrained using the data from both day 1 and day 2; 2- modified feedback was provided.

The same paradigm as on days 1 and 2 was used but 10 participants were provided with modified feedback (MDF) of their performance, whereas 10 participants were provided with feedback reflecting their actual performance (regular feedback, RGF). Participants were randomly assigned to one of the two groups and were not informed of the possibility of altered feedback. In the MDF condition, the participants first performed two sets where the feedback of the BCI (the avatar's movements) was congruent with the classification of the participant's brain activity. This was followed by six sets of positive MDF, wherein the feedback of the BCI was set to reflect the cued movement, regardless of the participant's brain activity, in 70% of the trials. In the remaining 30% of trials in sets 3 through 8, and in all trials of sets 9 and 10, the feedback of the BCI reflected the classification of brain activity.

2.2.3 Multi-Modal Self-Paced BCI design

The main goal of day 4 was to control a self-paced BCI in two randomly presented modalities: continuous and switch modes. Before this session, the two classifiers were updated and retrained again using the data from days 1, 2 and 3. Participants completed 10 sets of 15 trials where they were presented one of the three action cues, in randomized order. They were informed that when they received a forward walking cue, their avatar would move forward as long as they maintained the imagination of this movement (continuous mode). When they received right and left step cues, their avatar would take a single step forward after each correct MI task (switch mode).

There were 3 requirements that had to be met for a trial to be considered successful. First, the participant's avatar had to reach a target arrow by either sustaining MI of walking forward or by imagining multiple successive steps (starting with a left or right step, depending on the received cue). The second requirement was for the target arrow to be reached within a time frame of 24 s. In the continuous control mode, it took 12 seconds of sustained MI to reach the arrow, starting from the time the avatar started moving. In the switch mode, six steps were necessary to complete the entire trajectory. The third requirement was for the participant to initiate movement within 5 s of the presentation of the cue and to not stop forward progression for more than 5 s. In other words, if the classifier was classifying no-movement for 5 consecutive s, that trial was ended, and the next trial started.

For forward walking trials, the participants were instructed to maintain the imagination of forward walking as long as possible or until the avatar had reached the arrow. If the participant stopped maintaining the imagination at any time during the trial, the movement of the avatar stopped. The system allowed the participant to restart the same movement if he again produced the correct imagination signals, as long he/she was within the 5- and 24-s timeframes.

For switch mode trials, the participants were instructed that they were free to control the avatar any way they wanted but that the best strategy was to alternate between left and right steps. For example, if the they received a right cue, the best strategy to move forward was to imagine a right step (according to the cue), then a left step, then a right one, and so on until they reached the arrow.

2.3 EEG Signal Pre-processing

The EEG was processed using Matlab/EEGLab. The signals were amplified, band-pass filtered (18th order butterworth IIR filter) between 8 and 30 Hz and sampled at 256 Hz. Artifacts and noise were automatically rejected using an automatic independent component analysis (ICA) and noise-rejection algorithm (MARA, a plugin of EEGLAB) and the signals were then detrended, where an automatic baseline removal algorithm was applied. The referenced baseline that was used was the

200ms segment right before preceding the apparition of the arrow.

2.4 Feature Extraction

When using MI of the feet, it is very important to select the most informative features that can be used to determine the specific brain areas and activity that differentiate idling (no movement), right steps, left steps and forward navigation. In this study, power spectral density (PSD) of every channel, and 5 PSD asymmetrical ratios (1 s hanning) were chosen as the feature sets. From the data of day 1, these features were segmented between 4 – 8 s using 200-ms epochs with no overlap, resulting in 20 different epochs. This was done in order to identify the optimum time slot (and thus feature points) where the imagery signal achieved the best classification accuracy [39]. Data were also segmented over the 8-30 Hz frequency range using 3 Hz bins with 2 Hz overlap, in order to investigate the optimum frequency range where the MI signal resulted in the best classification performance [39]. This resulted in a total of 10 frequency bins and, thus, in 200 feature sets. The PSDs in each specific segment and frequency range were all concatenated together to form the feature set.

2.5 Feature Selection

The number of features was too high compared to the number of samples, which is known to lead to the existence of noisy features in the feature set and to longer processing time, reducing BCI efficiency. Wilcoxon signed rank based feature selection was used to remove redundant features [74]. The mean absolute value of the cross-correlation coefficient was calculated between each feature and all other individual features. The features that were not significantly different ($p<0.05$) from all existing features were removed.The result of this processing stage was a smaller dataset with a further distinguished set of features, for a better classification.

2.6 Classification

The selected features were fed into 2 regularized linear discriminate analysis (RLDA) classifiers where a 10-fold cross-validation was used. Classifier 1 (Cl1) was trained to distinguish between forward walking and no movement. Classifier 2 (Cl2) was trained to distinguish right step, left step and no movement. On day 4, the two classifiers were employed together. The EEG signal was first classified by Cl1. If it was classified as no movement, the signal was then classified by Cl2.

RLDA is a classification method that has been used in many MI-BCI studies [76]. Since RLDA is a regularization technique, it is particularly useful when there is a large set of features. The regularization amount hyperparameter λ is used to enhance the model by omitting predictors without decreasing the predictive power of the model. Thus, the regularization improves the classification performance by:

1) stabilizing the variance
2) reducing the bias of the discriminant function
3) providing generalization
4) and thus, preventing overfit
5) providing high robustness with respect to outliers
6) decreasing calculation time compared to other classification methods [77].

The amount of regularization was optimized by incrementing hyperparameter λ by 0.1 over the range of [0, 1.0] in a random search. It was validated by using cross-validation. The two RLDA classifiers were trained using the optimal λ.

2.7 Performance Evaluation

The performance of the cue paced BCI (days 2 and 3) was evaluated by the number of correctly classified trials. The performance of the self-paced BCI (day 4) was evaluated using the following parameters:

1. BCI performance accuracy: for each trial, if the participant met all 3 requirements, it was considered successful.
2. Trajectory completion time: the average time it took to complete the trial.
3. Average Number of stops: during forward walking, the number of times a participant stopped before reaching the target.
4. Average Stop times: the mean time elapsed between each step, in switch mode.
5. Steps alternation performance: the number of times a participant was able to alternate between left and right steps, in switch mode.
6. Maximum walk-maintain time: the longest amount of time the participant was able to maintain the MI of forward walking, without any stops or interruptions.

All parameters were averaged over the last two runs.

2.8 Spectral Map Analysis

Spectral power analyses were performed over only the segments and frequency bands that yielded the best classification accuracy. To verify the significant difference of EEG spectra over pair-wise comparisons, the paired t-test was performed. Permutation statistics was used as well, and the p-value threshold was set to be 0.05. Then, for multiple comparisons, a false discovery rate (FDR) was applied.

2.9 Questionnaires

After each day of training, participants were asked to answer a questionnaire on a computer. The questionnaire consisted of 18 questions on a 7-point Likert scale with: strongly disagree (1), disagree (2), somewhat disagree (3), neither agree nor disagree (4), somewhat agree (5), agree (6), strongly agree (7). The questions were an evaluation of the sense of presence (1-3), body ownership (4-6), sense of agency (7-9) , BCI performance (10-12), BCI tasks and design (13-15) and BCI environment feedback (16-18) [78]. Participants were able to complete the questionnaire in approximately one minute. Each embodiment component was rated with 3 questions, and each of the aspects of the BCI system was also rated with 3 questions. The score for each component was calculated as the average of the three questions [42]. A stronger sensation for a given aspect or component would be reflected by a higher score (Figure 10).

The questions were:

Presence

1. I had the feeling that the projection of the avatar was of my body
2. I felt like I was in a hallway
3. I forgot my presence in the real world, believing that I was in the virtual corridor, and that the avatar was me

Ownership

4. I felt like the virtual leg was my own leg
5. I felt like the avatar was me and not just a picture
6. When the virtual leg was moved, I felt that my own leg was moving, as if I was walking

Agency

7. The movements of the avatar corresponded to my thoughts/movements
8. The movements of the avatar were caused by my thoughts/movements
9. I felt that I was walking with my step and not with someone else's step

BCI Performance

10. My performance in controlling the walk of the avatar was mostly good
11. I felt disappointed with the avatar following my orders
12. I felt confident with the imaginative strategy of my feet

BCI Tasks

13. Controlling the movement of the left foot was easy
14. Controlling the movement of the right foot was easy
15. Controlling the forward walking movement was easy

BCI Feedback

16. I was comfortable with the starting arrows
17. I was comfortable with the avatar during the control of his walk
18. I was comfortable with the virtual corridor during control of the avatar's walk

2.10 Statistical Analysis

The Shapiro-Wilk test was performed to verify normality of all the collected data. For all statistical analysis over multiple days, or over more than two groups in the same condition (online and offline classifier performance with different retraining; subjective questionnaires) the Shapiro-Wilk test revealed that the normality assumption was not rejected. Thus, two-ways repeated measures ANOVAs were used. Pairwise comparisons were performed using Tukey HSD post-hoc test. For statistical analysis between two groups, the Wilcoxon test was performed over all paired samples when the normality assumption was rejected for at least one of the two groups (this was the case for the success rate and the performance parameters of the self-paced BCI). When the normality assumption was not rejected (this was the case for spectral power comparisons), parametric paired t-tests were used. In all cases, the threshold significance level was set to 0.05. Statistical significance is indicated in the figures by: * when $p < 0.05$, ** when $p < 0.01$ and *** when $p < 0.001$.

3. Results

For all results presented, the classification was performed over the features that yielded the best classification results. For frequency bands, these were sensorimotor rhythms (SMR) (12-15 Hz) for 2 participants and upper μ (10-13 Hz) for the other participants. For the segments, these were the 5th epoch for 3 participants, and the 3rd epoch for the rest of the participants.

3.1 Retraining of the Classifiers – Cue-Paced BCI

The first technique implemented to enhance the BCI results was to retrain the BCI after every session. The results show a mean classification accuracy of 71% on day 1 that is increased to a mean classification accuracy of 90% on day 3, when retraining was done using days 1 through 3. Repeated-measures ANOVA over retraining through days yielded a statistically significant effect of "Retraining" on performance on the 2 days of training (F=18.26, $p < 0.001$) and a significant effect of "Days" x "Retraining" interaction (F=9.02, $p < 0.05$). The results of pairwise comparisons over retraining revealed that the classification performance was significantly increased on: day 2 when training classifiers on data from days 1 + 2 versus data from only day 2 (diff = 13.77, $p < 0.01$); day 3 when training classifiers on data from days 1 + 2 + 3 versus data from days 1 + 3 (diff = 8.38, $p < 0.05$) or days 2 + 3 (diff = 8.45, $p < 0.05$); day 3 when training classifiers on data from days 1+ 2 + 3 versus data from only day 3 (diff = 14.83, $p < 0.01$). No significant differences were found between training with day 3 only and training with days 1 + 2. The results of pairwise comparisons over retraining and days revealed that the classification performance was significantly increased on day 2 when training classifiers on data from days 1 + 2 versus data from only day 1 (diff = 12.9, $p < 0.01$) and on day 3 when training classifiers on data from days 1 + 2 + 3 versus data from day 1 only (diff = 18.84, $p < 0.001$). No significant differences were found between day 1 only and day 3 only, nor between day 2 only and day 3 only. Figure 5 shows the BCI performance of the offline classifiers, trained with the data from different combinations of days.

Figure 6 illustrates the effects of the retraining technique over online performance of the cue-paced BCI. It shows the online performance at the end of day 2 and the online performance at the beginning of day 3, i.e. after retraining the classifiers with the data of day 1 and day 2. The average online performance at the end of day 2 (pre-retraining) was 65.8%, which was increased to 74.7% at the beginning of day 3 (post-retraining). Repeated measures ANOVA over retraining and classifiers through the different days yielded a statistically significant effect of "Retraining" (F=19.32, $p < 0.05$) and a significant effect of "Classifiers" (F=24.64, $p < 0.001$). No significant effect of "Classifiers" x "Retraining" interaction was found. The results of pairwise comparisons over "Retraining" revealed that, for Cl1, performance was increased significantly from 58.6±9.5% (pre-retraining) to 70.7±5.5% (post-retraining) with (diff = 12.10, $p < 0.05$). For Cl2, performance was increased from 73.0±12.2% (pre-retraining) to 79.01±11.15% (post-retraining), which was not statistically significant. In pairwise cross-comparisons between classifiers within the group, a significant difference was obtained between Cl1 and Cl2 in pre-retraining (diff = 14.43, $p < 0.001$) and post-retraining (diff = 8.30, $p < 0.05$).

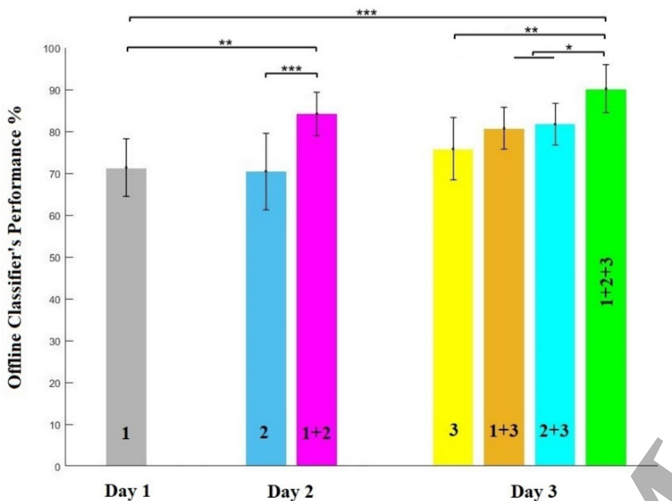

Figure 5: Offline Classification results averaged over the 2 classifiers (Cl1 and Cl2) for the first 3 days of training. It shows the mean accuracy of the offline classifiers on each day, when trained using data from that day only and when trained using data from different combinations of days (the numbers on the columns indicates which days were used for training).

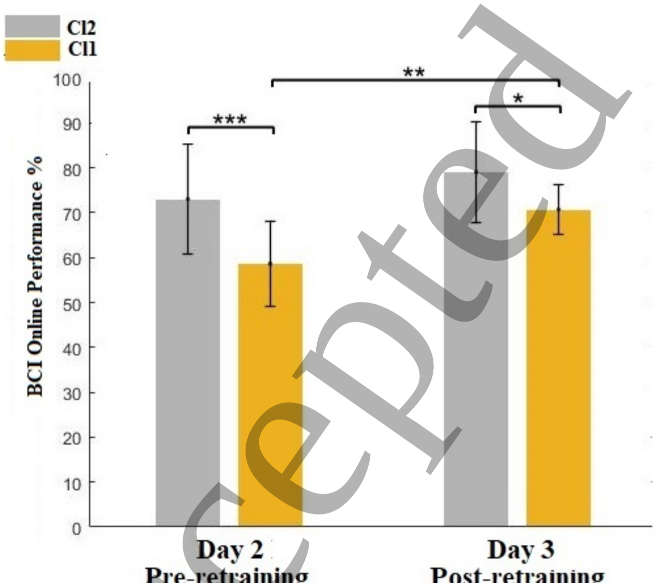

Figure 6 Online performance results at the end of day 2 and at the beginning of day 3, for each of the 2 classifiers.

3.2 Modified feedback – Cue-Paced BCI

The average online performance (across both classifiers) at the beginning of the training on day 3 was 74.7% (pre). At the end of the training day, it reached 75.5% for the group that received regular feedback (post-RGF) and 85.9% for the group that received modified feedback (post-MDF). Figure 7 shows the online BCI performance for each of the 2 classifiers in the first two and last two runs of day 3. The performances for the last 2 runs are separated by group (RGF vs MDF).

Repeated measures ANOVA over Feedback and Classifiers for performance in day 3, yielded a statistically significant intra-subject main effect of "Feedback" ($F=37.04$, $p < 0.001$) and a significant effect of "Classifiers" ($F=17.22$, $p<0.001$). The effect of "Feedback" x "Classifiers" interaction was not significant. The results of pairwise comparisons over feedback revealed that for the MDF group, Cl1 performance increased significantly from 70.7±5.5% at the beginning of the training in day 3 (pre), to 82.1±4.9% in post-MDF (diff = 11.39, $p < 0.001$). There was a significant difference of Cl1 between Post-MDF and post-RGF (diff = 11.12, $p < 0.01$). The performance of Cl2 increased significantly from 79.0±11.2% in the first runs of day 3 (pre) to 89.7±2.9% in post-MDF (diff = 10.68, $p < 0.01$). There was a significant difference of Cl2 between post-MDF and post-RGF (diff = 9.28, $p < 0.01$). In pairwise cross-comparisons between classifiers within the same group, a significant difference was obtained between Cl1 and Cl2 in post-MDF (diff = 7.59, $p < 0.001$) and post-RGF (diff = 8.87, $p < 0.05$). No significant difference between classifiers was found in the first 2 runs. Overall, providing 6 sets of positive MDF enhanced the BCI's online performance significantly.

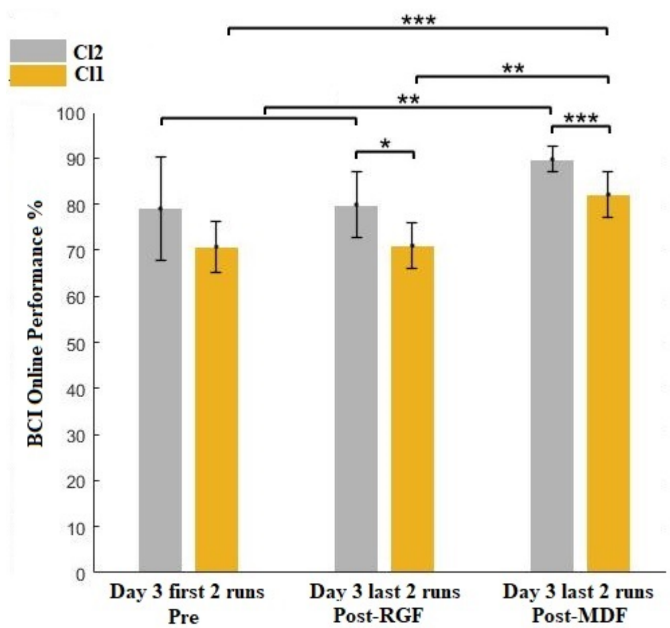

Figure 7: Online Classification of the 2 classifiers, averaged over the first 2 runs of day 3 and the last 2 runs of day 3, with and without modified feedback (MDF).

3.3 Modified feedback- Cue-Paced BCI : Neurophyiological Results

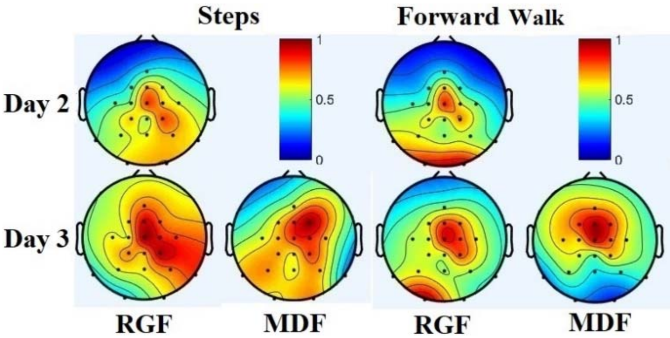

Figure 8: Spectral power maps (10*log μv^2/Hz) of the upper μ frequency band (10-13 Hz) over the epoch that yielded the best classification accuracy, over day 2 and day 3, and for the MDF and RGF condition for the cue-paced BCI. The results revealed a strong central and parietal ERS in the case of RGF, compared to a stronger frontal ERS in the case of MDF.

Using the segments and frequency bands that yielded the best classification accuracy, spectral power maps were plotted (Figure 8). During analysis, the EEG spectral topography revealed a bilateral inversion of the MI left and right steps. Thus, in order to facilitate the analysis, these maps were mirrored and merged into a single map.

The analysis revealed significant differences between trials with RGF and MDF. When controlling the steps of the avatar, one can observe an activation over the central and parietal areas of the brain, which is getting stronger at the end of the training. When MDF is provided, these spectral powers increase over the frontal areas, with a significant difference ($p < 0.05$). When controlling the forward walking of the avatar, central power peaks are observed, spanning more to frontal and parietal areas at the end of the training. However, when MDF is provided, these spectral powers increase over the frontal areas, with a significant difference ($p < 0.05$).

3.4 Self-Paced BCI

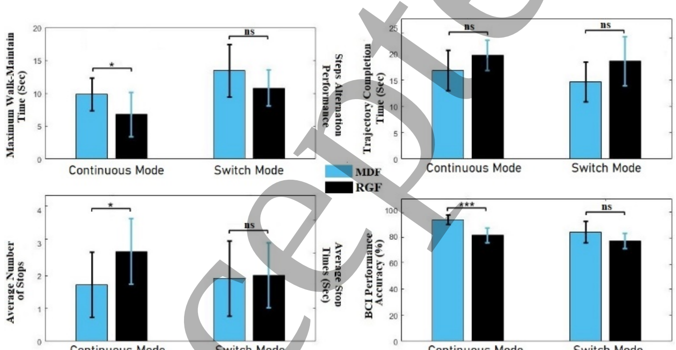

Figure 9: Online classification results for both self-paced BCI modes, for participants who received regular feedback (RGF, in blue) and modified feedback (MDF, in black). Each of the 4 plots depicts a parameter of evaluation of the performance of this BCI, which are in order: Maximum walk-maintain time/ Steps alternation performance, Trajectory completion time, Average Number of stops/ Average Stop times and BCI performance accuracy.

In continuous mode, BCI performance accuracy was significantly increased ($Z=3.4395$, $p < 0.001$) in the MDF group (92.5 ± 3.5% success rate) compared to the RGF group (80.8 ± 5.6% success rate). Maximum walk-maintain time was also increased significantly to 9.6 ± 2.4 s with MDF versus 6.8 ± 3.3 s with RGF ($Z=2.0301$, $p < 0.05$). The average number of stops was significantly decreased from 2.7 ± 1 stops with RGF to 1.0 ± 0.9 stops with MDF ($Z=-2.0062$, $p < 0.05$). Trajectory completion time was decreased from 19.7 ± 2.8 s with RGF to 16.8 ± 3.8 s with MDF. Trajectory completion time was not significantly different between groups.

In switch mode, results show a tendency towards improved BCI performance accuracy with MDF but differences between groups were not statistically significant. BCI performance accuracy was increased in the MDF group (82.9 ± 8.1% success rate) compared to the RGF group (76.5 ± 5.8% success rate). Step alternation performance was increased to 67.1 ± 20.0% success rate with MDF, from 53.9 ±13.0% success rate with RGF. The average stop times between steps was decreased from 2.0 ± 1.9 s with RGF to 1.9 ±1.1 s with MDF. Trajectory completion time was decreased from 18.6 ± 4.7 s with RGF to 14.6 ± 3.7 s with MDF.

3.5 Behavioral Results

Repeated measures ANOVA were run over the 3 embodiment components and 3 BCI design aspects. It revealed a significant intra-subject main effect of ‘Training’, a significant effect of ‘Feedback’. Pairwise comparisons revealed that when operating a BCI to control the gait of an avatar, the sense of presence, which starts at 5.4/7 on day 1, does not significantly change through the days, with no significant effect of MDF.

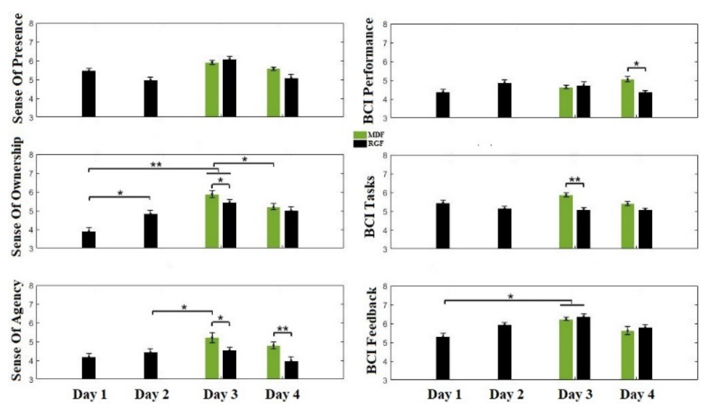

Figure 10: Behavioral results in the form of the participants’ answers to the questionnaires. Those questionnaires were answered by the participants after each day of training. The three plots on the left represent the 3 embodiment components, where the three plots on the right represent the 3 BCI aspects and tasks.

The perceived body ownership, which starts at 3.9/7 in day 1, increased significantly to 4.8/7 in day 2 (diff = 0.93, $p < 0.05$). Ownership also increased significantly from day 2 to day 3 for both the MDF (diff = 1.96, $p < 0.01$) and RGF (diff = 1.53,

$p < 0.01$) groups. For group RGF, it decreased significantly from day 3 to day 4 (diff = -0.92, $p < 0.05$). The MDF group had no significant difference between days 3 and 4. In day 3, ownership was significantly higher in the MDF group than in the RGF group (diff = 0.42, $p < 0.05$).

The perceived sense of agency, which starts at 4.2/7 in day 1, doesn't significantly change through the days for the RGF group. In the MDF group, it increased significantly from 4.4/7 in day 2 to 5.2/7 at day 3 (diff = 0.78, $p < 0.05$). The sense of agency was significantly higher in the MDF group (vs RGF group) on day 3 (diff = 0.74, $p < 0.05$) and day 4 (diff = 1.01, $p < 0.01$).

As for BCI design, pairwise comparisons show that the perceived feedback of the cue-paced BCI was increased significantly from day 1 to day 2 in the MDF group (diff = 1.06, $p < 0.05$) and in the RGF group (diff = 0.93, $p < 0.05$). Compared to the RGF group, the MDF group had significantly higher perceived performance with the self-paced BCI ($p < 0.05$) and significantly higher perceived easiness of tasks with the cue-paced BCI ($p < 0.01$). There was no significant difference between groups in BCI environment feedback questions.

4. Discussion

This study aimed to develop a gait rehabilitation BCI, integrating MI of lower limbs to control the gait of a virtual avatar in different control modes, such as cue-based and self-paced. This approach was used in order to overcome the design limitations of the currently used gait rehabilitation BCIs. This study also investigated the implementation and combination of different BCI performance enhancement techniques such as co-adaptive sequential training paradigm, modified feedback and classifier retraining. This was a novel approach for lower-limbs MI-BCIs, because, to our knowledge, there is no previous study that integrated all of these techniques in one single BCI system that use MI of lower limbs. It was used in order to overcome the performance limitations of such BCIs that would hinder embodiment and limit use in gait rehabilitation. The results confirm our hypotheses that participants were able to operate the BCI in all modes, and that the three enhancement techniques increased BCI performance. However, the hypothesis that participants would perform better when using the continuous self-paced control than when using switch-based self-paced mode is rejected.

4.1 Retraining of the Classifiers – Cue-Paced BCI

The offline classification results show that on day 1 of the training, the classifiers reached a performance that was at least 10% higher than the chance level, in their worst case. This classification accuracy was enhanced significantly after using the recalibration technique. When the trained classifiers were used online, they were able to perform with a strong generalization. This was clear from the online results of day 2. The recalibration technique was shown to perform well, not only for offline classification, but also for online classification. This is shown by comparing the online performance at the end of day 2, when the classifiers were trained on data from day 1 only, with the online performance at the beginning of training on day 3, where the recalibration technique enhanced the BCI performance by 10%. This is similar to the range of enhancement (10-15%) found by other studies [71-73, 79]. The classification accuracy was significantly enhanced for Cl2 and for the combination of both classifiers. It is worth noting that the results always show a higher classification accuracy for Cl1 versus Cl2. This lower performance and higher enhancement for Cl2 may be due to the complexity of this MI tasks (left step, right step or no movement) and the fact that there are 3 classes, compared to the two tasks classified by Cl1 (walking forward or no movement). Nevertheless, the classifiers reached high classification performance for many participants. After retraining, they achieved offline classification accuracies at day 3 between 70% and 95% (mean: 86%), which were significantly superior to chance.

In the field of BCI, the offline classification accuracy for a MI-BCI is considered to be good if it reaches more than 70% , which is the case in this study, even before retraining the classifiers. Furthermore, the findings after retraining were consistent with what other BCI studies have previously found [68, 69], e.g. that, at the stage of classifier design, using an ensemble of LDA with regularization and retraining of classifiers after every session provided high classification accuracy and performance [79]. The results of the classifications of forward walking in this study were better than the performances that were reported in studies that used the RLDA algorithm to classify walk vs no_move, which were between 70 and 80% [80, 81]. This could be due to the technique of retraining the classifiers used in this study. To our knowledge, this is the first study that has used RLDA classification for discrimination of right step, left step and no movement. This is also the first study to integrate RLDA with the classifier retraining enhancement technique for a lower-limb MI-BCI. Our findings demonstrate the feasibility of using these methods for such BCIs.

The high classification performances presented in this study also support the feasibility of using the features of PSD and PSD asymmetrical ratios between the two brain hemispheres for encoding the differences between the main control commands of gait. This is in accordance with what was found in previous studies that investigated EEG signatures of gait [43, 45, 47]. On the other hand, the optimized feature sets had data only from the frontal, prefrontal and central areas of the brain. Since those channels are localized over the foot representation areas of the brain, it could be possible to further lower the number of electrodes from 19 to 10 or less.

4.2 Modified feedback – Cue-Paced BCI

The results show that providing 6 runs of positive MDF enhanced the cue-paced BCI performance significantly when compared to the group that didn't receive MDF. The results show that with MDF, the average general performance reached ~83%, an accuracy that is considered to be very good in the field of MI-BCIs [82]. The positive MDF used in this study enhanced the mean performance over both classifiers by 10%, with a 12% enhancement for Cl2 specifically. When previous studies compared the effects of providing positive and negative MDF, some found that it was negative MDF that had the larger enhancing effect on BCI performance. This was the case in studies that trained users to control a bar on the screen [63, 65]. Others found that positive MDF had a larger effect on enhancing BCI performance. This was the case, for example, of a study that trained users to control the hand of human-like robot [66]. The results found by the latter are consistent with the findings of our study. The different findings with regards to the superiority of positive or negative MDF may be due to the nature of the study and, specifically, to the nature of the feedback to be controlled. Since this is, to our knowledge, the first study to use MDF to enhance the performance of a multi-command lower-limb MI-BCI, the results show the feasibility of using this method to enhance such BCIs.

MDF also contributed to the reorganization of cortical activation after participants trained to control the BCI. Spectral power maps show that when controlling the forward walk of the avatar, a stronger activation is observed over the central areas. However, when controlling steps, a stronger activation is observed over central and parietal areas of the brain. One can also observe a stronger activation at the end of the training, especially at the right-central regions. These brain activations can be described as large event-related synchronisations (ERS). ERS usually occur as signal rebounds after a mental activity, and they are characterized as an increase of the signal power, compared to baseline. In this study, after MDF was provided during gait control, these ERS peaks could be due to the higher feeling of agency [42].

The central and right-central μ-ERS may have been due to the role of MI mechanisms over the brain areas of the feet. This implies an elevated degree of the sense of ownership when controlling the limb by IM. The parietal μ-ERS could be explained by 2 factors. First, this is the region of the brain implicated in dimensional navigation, and thus, this is the region that is activated when the feeling of presence takes place. Second, the angular gyrus is situated in this region and in the inferior parietal cortex in particular. This region is involved in the feeling of agency and the operation of comparison between the anticipated and real outcomes of one's actions [83, 84].

When MDF was provided, these spectral powers increased significantly to a strong parietal, central-frontal and frontal μ-ERS. The frontal activation represents the judgement of agency, which suggests that the perceived feeling of agency was significantly higher in the group that received MDF [85]. The stronger frontal central sensorimotor rhythm (SMR) and μ-ERS may explained by one of the following:

1- When the participant wants to either hold or end the movement upon reaching the target position, the elevated process of the mismatch-surveillance circuits of the brain and the compensatory neural control mechanisms emulate this type of brain activation [86]. It was found that for the group that didn't receive MDF, this specific activation was much lower, and it could be a consequence of the regular process of the mismatch surveillance circuits of the brain.
2- The mirror neurons system (MNS) situated in the frontal cortex. This system is exhibits specific brain activation that tunes and modulates μ-ERS to the goal-directed motor experiences in movement imitation of the avatar (frontal). This activity follows movement observation of the avatar (central). These activations are necessary for the brain to comprehend and handle actions of the avatar [87].
3- The working memory consolidation [88]. Because the sequential training to control the BCI presented in this study lasts four days, the working memory could have been consolidated over these days. This means, that the skills that the participants have acquired during the training have accumulated during these days. Thus, an enhanced MI performance could be obtained.

Remarkably, when MDF was not provided, the frontal-central and parietal μ-ERS were higher for the control of steps than for the control of forward walking. This difference can be explained by the different complex compensatory neuronal control circuits that were triggered right after a movement of the avatar that conflicts with the participant's MI, in both conditions. In this study, the movement violation that resulting from a misclassification of that command was delivered as an absence of feedback, in the case of forward walking. In the case of individual steps, it was an absence of feedback or a contradictory feedback. The results show that the second demands more power from the brain because it has to where suppress the anti-saccadic effects. Thus, when MDF was not presented, this triggered a stronger central frontal activation in the individual steps condition. This finding is consistent with the results of previous studies that showed that for an anti-saccadic movement, the neural and perceptual consequences are stronger than those of a movement inhibition [89, 90]. This also explains the apparition of significant post-MDF changes over the parietal μ-ERS.

These power spectral maps findings on the effects of MDF on MI are consistent with the results of previous studies [42]. To summarise, the results show that when mentally controlling the gait of an avatar, MDF enhances cue-paced BCI performance, especially when controlling harder tasks, such as right and left steps. This performance can be measured by the classification rate and by the μ-ERS power spectral changes over the central-frontal and the central-parietal areas.

4.3 Co-adaptive sequential training - Self-paced BCI

To reach high performance in controlling the self-paced BCI in its two modes, an effective, goal directed and short-time sequential training for operating a cue-based BCI is needed. Participants gained good control of the self-paced BCI after being trained for 3 days to control the cue-paced BCI and after having completed only a few runs of controlling the self-paced BCI. They were able to control the steps and forward walking of their self-avatar after only 4 days of training, in total. Other studies have used the co-adaptive sequential training paradigm to train users to control a cue-paced followed by a self-paced

BCI for cursor control , avatar control or navigation in VR [8]. The required time for this training varied between a few sessions in one day and many sessions over several days [61]. For gait rehabilitation MI-BCIs, some participants requires weeks of training to control a self-paced BCI [51].

The short amount of training time that was sufficient in this study may be due to the three BCI enhancement techniques that were combined and implemented throughout the training. To our knowledge, this is the first study that investigated the feasibility of using a combination of all of these enhancement techniques, in the same training paradigm, in order to control a lower-limb MI-BCI. Our results suggest that such a method could potentially shorten the training time required to reach a high performance in controlling the cue-paced, and later the self-paced BCI.

The general performance scores in both modes of our self-paced BCI were 85% in average. This is equal to or higher than what was found in BCI-VR studies to control a cursor , upper limbs or even gait [4]. Considering the complexity of the tasks being carried out in the current study, and compared to other rehabilitation studies, the performance is satisfactory.

Some of the parameters used to evaluate performance of the self-paced BCI, such as step alternation, were specific to our study and therefore cannot be directly compared to existing literature. Other parameters, such as 'Maximum time maintaining the MI' and 'stop times' are at the core of self-paced BCIs and have been reported in previous studies. In the self-paced continuous control mode, participants were able to maintain the MI for a long enough time to reach the target with minimal or no interruptions (mean of 1.0 stops for MDF group and 2.7 stop for RGF group). As for the self-paced switch control mode, participants were able to alternate their steps, and reach the target with short times intervals between steps (mean of 2.0 s between steps for the MDF group, 1.9 s for the RGF group). This is consistent with the findings of previous studies, which found that users were able to reach a high level of performance in evaluation parameters when using different control modes of a self-paced BCI [8, 54, 91].

Scores were generally high for the different performance parameters in both modes. However, it was surprising to observe that the effects of the positive MDF, provided for the cue-paced BCI on day 3, had spanned to significantly increase the score of most of the parameters that were used to evaluate the performance of the continuous self-paced BCI on day 4. Participants from the MDF group may have been more confident in their results and performance, more concentrated and more motivated to control the avatar and finish the training successfully. In comparison, participants in the RGF, may have felt more discouragement or frustration with the BCI after non-successful trials. Other hypotheses are that differences could be related to memory consolidation or that they were due to cortical activation re-organization and the MNS modulation that followed MDF training. A combination of these factors may be in play. To our knowledge, this is the first study to show that MDF can improve BCI performance in following training sessions, on subsequent days. Alimardani et al. , using an upper-limb MI-BCI, found that the improvement in participant's performance following MDF carried over to subsequent training sessions of a same day. The effect on sessions that occurred on different days was not investigated. Overall, our results and those of Alimardani et al. show that MDF can be used in sequential training MI-BCIs for the objective of increasing the performance of participant's in controlling the BCI.

On the other hand, MDF did not have the same beneficial effect on performance in the switch-control mode. The scores in the different performance parameters were improved but not significantly. In the continuous control mode, the participants had to maintain the MI for a longer period of time, compared to the switch mode. Although the switch control mode is considered to require less concentration, it consumes more brain-power due to the anti-saccadic mechanisms. Thus, participants performed better using in continuous control mode. It is well known that walking requires a very complex sequence of upper and lower limbs movements. Consequently, there may not exist a single common mental strategy of gait motor imagery strategy. Thus, participants may have been employing various mental strategies for the same task, which may explain those differences. This is consistent with what Valesco-Alvarez et al. found when comparing the two self-paced BCI modes in order to control navigation in VR. In their study, the success rates were similar to ours for both control modes, although the times needed to complete a path were notably lower in the continuous control mode.

To summarise, these analyses suggest that a sequential co-adaptive training to control a cue-paced, followed by a self-paced MI-BCI is very successful. The implementation of an ensemble of the previously mentioned enhancement techniques may shorten the training time. The implementation of MDF technique specifically also enhanced the performance, especially for more difficult tasks.

4.4 Behavioral measures of embodiment and performance

In order to operate a BCI that controls the gait of an avatar, there is a very strong link between the degree of embodiment over that avatar, and the level of performance to control that avatar. To reach one of them, the second must be reached as well [23]. In this study the feedback of MI to control the gait of an avatar resulted in elevated levels of embodiment of the 1PP avatar, which is consistent with previous literature [92, 93]. In a previous study, the effect of MDF over the central-frontal and the parietal areas correlated with the subjective evaluation of embodiment [94]. That means that the ERS modulations following agency violations were stronger with participants who experienced stronger embodiment, which justifies the use of questionnaires in this experiment.

The scores of the questionnaires revealed that the sense of presence was generally high (> 4.9/7), especially in day 3 when participants were mastering the control of the cue-paced BCI, whether they received MDF or not. This is also consistent with what was shown in the spectral power maps, regarding the parietal μ-ERS that peaks when there is a high sense of presence. The perceived sense of ownership was low at the beginning of the training but was increased significantly at the end of the training to control the cue paced BCI (day 3). This is consistent with the finding that the right-central peaks were stronger at the end of the training. MDF significantly increased the sense of ownership on day 3 but not when controlling the self-paced BCI (day 4). Again, this may be due to the avatar's behavioral differences between a cue-paced control and a self-paced control BCI. The sense of agency was low at the beginning of the training but MDF participants felt significantly more in control (agency) on day 3 and on day 4. This is consistent with what was found in the neurophysiological results: when producing MDF to the gait of a virtual avatar, the sense of agency can be measured using the strength of the elicited frontal central ERS.

The embodiment question scores showed that participants felt immersed, which was confirmed and supported by the results of previous studies. Thus, when an avatar's gait is presented to the user from 1PP in an immersive VE, the feedback of MI, in the form of the avatar walking, is sufficient to produce the sense of embodiment [94]. This was confirmed as well by the high scores that were observed in the answers of the participants to the questions related to the BCI feedback, which reached 5.3/7 in day 1 and got increased to to 6.3/7 in day 3.

The BCI question scores showed a high perceived performance and easiness of tasks for the MDF group, when controlling the self-paced BCI, with a higher score of tasks for the MDF groups when controlling cue-paced BCI. MDF probably played a strong role on perceived performance, which increased agency and embodiment. With positive modified feedback (MDF group), the feedback is more often consistent with what the participants are trying to imagine. Consequently, their performance perception and agency perception were increased.

Since the results of behavioral measures were higher for the MDF group than for the RGF group, the use of MDF during the self-paced BCI could probably improve the behavioral measures of the sense of agency, BCI performance and BCI easiness of tasks.

To summarise, these analyses suggest that when using a lower-limb multi-modal MI-BCI to control the gait of an immersive avatar, the questionnaire score results are in accordance with neurophysiological and performance measures and can be used as an assistive measure to monitor embodiment and performance.

4.5 General discussion and limitations

In general, when compared to related cue-paced or self-paced BCI-VR studies that control gait, the current study shows similar or better performances, despite variations in experimental designs. Our results show that the developed BCI could meet the requirements of an ideal BCI to control the gait of an avatar. The key factors that make BCIs a real alternative communication channel are: fast setup, short training time, effective control and artifacts processing [95]. In this study, some of these factors, such as short training used and the high performance the results have shown, were attained.

This study had some limitations. For one, many MI-BCI studies a classifier progress bar added to the VE, either separately on the screen, or super-imposed over the cue. This usually helps the user, in real time, to be aware of his performance in controlling the BCI and to try to adjust accordingly. The absence of such an aspect, left the participants sometimes unaware of the best mental strategy to use, especially when there were multiple mental commands at a time, such as in the switch control mode. Therefore, the absence this bar may have diminished the BCI performance achieved in this study, especially in self-paced switch mode. Had such a bar been present and improved performance, it could have presumably improved the behavioral measures of BCI performance, BCI tasks and sense of agency.

An important limitation of the setup was the use of generic animations of the avatar's gait. Several participants noticed and commented that the avatar's gait looked different from their own gait. Since this study is a first step towards a BCI that could be used for gait rehabilitation, it would be useful to have calibrated subject-specific avatar gait animations for future work. Such personalised animations could increase the behavioral measures of the sense of ownership and the BCI feedback.

Interestingly, the scores obtained in the subjective questionnaires could have reflected another limitation of this study. This experiment didn't use a gender-matched avatar contrary to what other studies have done. Non gender-matched avatars have been shown to decrease embodiment [33] and some of our female participants did question why they were embodying a male avatar. This could partly explain the low scores in the ownership questions in the first days of training, when the feeling of agency is less established.

5. Conclusion

This study reports on the feasibility and successful design and development of a BCI system that uses lower-limb MI to control the steps and forward walking of a self-avatar in immersive VR. Twenty participants were able to operate the BCI after 4 sessions co-adaptive sequential training of 1 hour per session, with a general performance of 70-94%. To our knowledge, this study represents the first demonstration of integrating all of these design approaches and enhancement techniques in parallel in one single multi-modal BCI system. This BCI could be used for gait rehabilitation by imagining real steps, and progressing to imagining forward walking, while receiving the matching visual feedback from an embodied virtual avatar over which we feel agency. Future work will incorporate proprioceptive feedback that is congruent with the movements of the avatar.

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